Astronomy & Astrophysics manuscript no. paper
December 19, 2025

Are galaxies in compact groups special?

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Received XXX; accepted YYY

ABSTRACT

We investigate the properties of galaxies in compact groups (CGs) and compare them to a control sample of galaxies.

Key words. galaxies: clusters: general – catalogs

1. Introduction

2. Data

2.1. Samples

We use the compact groups and control samples we built in our previous article Tricottet et al. (2025). We neververtheless briefly recall here how they were built. Explain. Contrarily to our previous article, though, we remove CGs that we classified as *split* following Zheng & Shen (2021). In this process, our initial compact group sample of 78 groups reduced to 62.

We use the SDSS DR 16 to retrieve masses, star formation rates (SFRs) and morphologies of galaxies assessed in Galaxy Zoo Lintott et al. (2011). We specifically extracted fields `sfr_tot_p50`, `specsfr_tot_p50` and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table.

2.2. sSFR

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 through the query displayed in listing 1.

```

    SELECT
        s.specObjID,
        s.z,
        p.petroMag_r,
        p.objID,
        g.sfr_tot_p50, g.specsfr_tot_p50, g.
            lgm_tot_p50,
        l.h_alpha_eqw, l.h_beta_eqw, l.
            oiii_5007_eqw, l.nii_6584_eqw,
        l.h_alpha_flux, l.h_beta_flux, l.
            oiii_5007_flux, l.nii_6584_flux,
        z. p_el_debiased AS p_E,
        z. p_cs_debiased     AS p_S
    FROM SpecObj AS s
        JOIN PhotoObj AS p ON s.bestObjID = p
            .objID
        JOIN galSpecExtra as g ON s.specObjID
            = g.specObjID
        JOIN galSpecLine as l ON s.specObjID
            = l.specObjID
        JOIN zooSpec AS z ON s.specObjID = z.
            specObjID
    WHERE s.z BETWEEN 0.005 AND 0.0452
        AND (p.petroMag_r - p.extinction_r <=
            17.77)
        AND s.class = 'GALAXY'
        AND g.lgm_tot_p50 > -1000
```

Listing 1: Query used for selecting spectral & photometric data from the SDSS

We select non-AGN galaxies by first placing them on the classical BPT diagnostic of Veilleux & Osterbrock (1987) and requiring measured values of both emission-line ratios

$$\log_{10}([\mathrm{N\,II}]\lambda6584/\mathrm{H}\alpha) \quad \text{and} \quad \log_{10}([\mathrm{O\,III}]\lambda5007/\mathrm{H}\beta).$$

A galaxy is flagged as an AGN if it satisfies either

$$\log_{10}\frac{[\mathrm{N\,II}]}{\mathrm{H}\alpha} > 0,$$

or if it lies above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10}\frac{[\mathrm{O\,III}]}{\mathrm{H}\beta} > \frac{0.61}{\log_{10}([\mathrm{N\,II}]/\mathrm{H}\alpha) - 0.05} + 1.3,$$

in which case it is removed from the non-AGN sample. All remaining galaxies—including those with missing line ratios—are retained as non-AGN. The non-AGN selection process is shown on figure 1.

We identify the galaxies having a `specsf_tot_p50` of -9999 in the SDSS as quenched (for graphic representation purposes, we give it here an arbitrary low value

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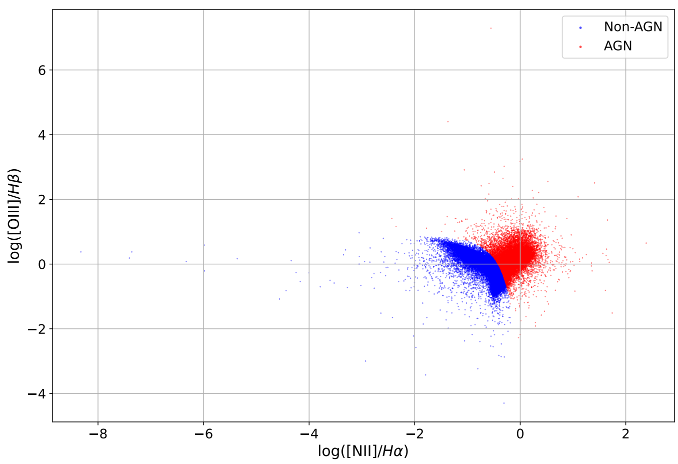

Fig. 1: BPT diagram showing our separation between ordinary and AGN galaxies

of $10^{-14.0}$ Gyr^{-1}). We then consider the other ones and sort them between the star-forming and passive populations using a two-component Gaussian Mixture Model (GMM) in the $\log M_*$–$\log$ sSFR plane. Its parameters determined by minimizing the Kullback–Leibler divergence between the empirical and model densities Kullback & Leibler (1951); McLachlan & Peel (2000). For any trial parameter vector $\boldsymbol{\theta}$, we reconstruct the mixture means $\{\boldsymbol{\mu}_i\}$, covariances $\{\boldsymbol{\Sigma}_i\}$, and weights $\{w_i\}$ via a Cholesky-like decomposition and logistic mapping, optionally constraining the second component's mean to isolate the passive population Dempster et al. (1977). The KL divergence is estimated by constructing a 2D histogram of the data and summing $p_{\rm data} \ln(p_{\rm data}/p_{\rm GMM})$, with a small regularization ϵ to avoid singularities Tojeiro et al. (2009); Bisigello et al. (2018). We optimize $\boldsymbol{\theta}$ via L-BFGS-B (with fallback to Nelder–Mead) over multiple guided and random initializations, selecting the solution with lowest divergence Pedregosa et al. (2011); Bishop (2006). Initial means are seeded by a median split in $\log$ sSFR to improve convergence and robustness against local minima Rousseeuw (1987). In the final fit, one component naturally aligns with the high-sSFR "star-forming sequence," while the other captures the low-sSFR "passive" cloud, providing a data-driven division that agrees with previous multi-Gaussian decompositions of the SFR–M_* plane Wuyts et al. (2011); Hahn et al. (2019). The decision boundaries, defined as the loci where the posterior probabilities of adjacent components are equal, provide an objective criterion for delineating star-forming from passive galaxies.

Add relevant GM parameters found by the algorithm.

The sSFR status for each sample is shown in table 1. Tere is no significant differences of for Star forming proportion between CG$_4$ and the control samples (probabilities estimated using Fisher exact test are 6.5×10^{-1} versus Control$_{4\rm B}$ sample, 2.5×10^{-1} versus Control$_{4\rm C}$ and 1.8×10^{-2} versus RG$_4$).

Restrictions of the comparison to BGGs and satellites only are also shown in table 1. The p-values found from two-sided Fisher exact test are 0.58 against Control$_{4\rm B}$, 0.59 against Control$_{4\rm C}$ and 0.37 against RG$_4$. There is thus no significant difference in the fraction of star-forming BGGs nor satellites between CG$_4$ and our control samples either.

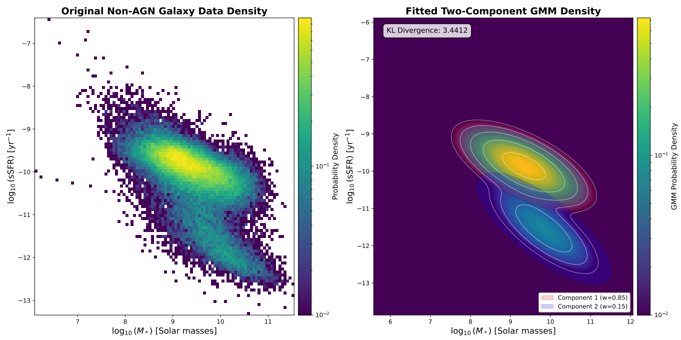

Fig. 2

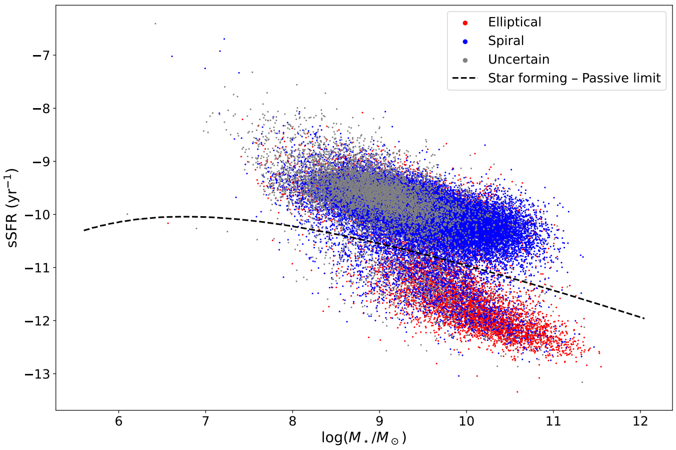

Fig. 3: sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

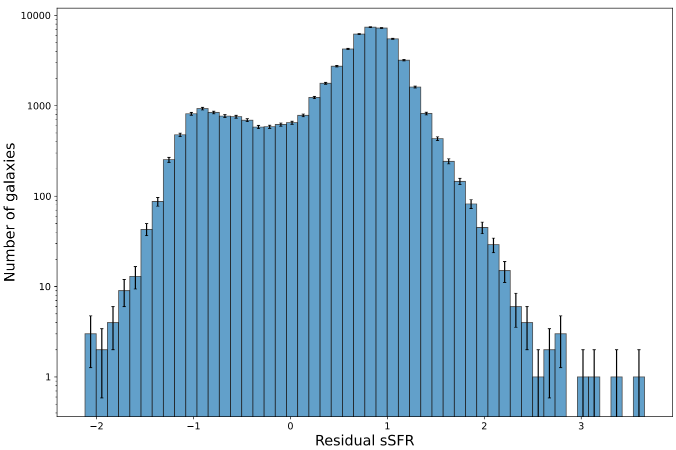

Fig. 4

2.3. Morphology

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiaised vote fractions Lintott et al. (2011). Those fractions are provided in the `zooSpecz` table of the SDSS databasethrough the `p_el_debiased` and `p_cs_debiased` columns. We attribute to galaxies the morphology "elliptical" if the `p_el_debiased` value is greater

Sample		Quenched	Passive	Starforming
CG_4		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	BGG	*6 (9.7 %)*	*45 (72.6 %)*	*11 (17.7 %)*
	Satellites	*12 (6.5 %)*	*101 (54.3 %)*	*73 (39.2 %)*
$Control_{4B}$		181 (6.5 %)	1595 (57.0 %)	1020 (36.5 %)
	BGG	*58 (8.3 %)*	*534 (76.4 %)*	*107 (15.3 %)*
	Satellites	*123 (5.9 %)*	*1061 (50.6 %)*	*913 (43.5 %)*
$Control_{4C}$		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	BGG	*65 (8.6 %)*	*569 (75.7 %)*	*118 (15.7 %)*
	Satellites	*126 (5.6 %)*	*1050 (46.5 %)*	*1080 (47.9 %)*
RG_4		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	BGG	*3 (5.4 %)*	*38 (67.9 %)*	*15 (26.8 %)*
	Satellites	*2 (1.2 %)*	*63 (37.5 %)*	*103 (61.3 %)*
SDSS selection		*634 (1.2 %)*	*7705 (14.7 %)*	*44192 (84.1 %)*

Table 1: Number of galaxies in each sSFR status for each sample.

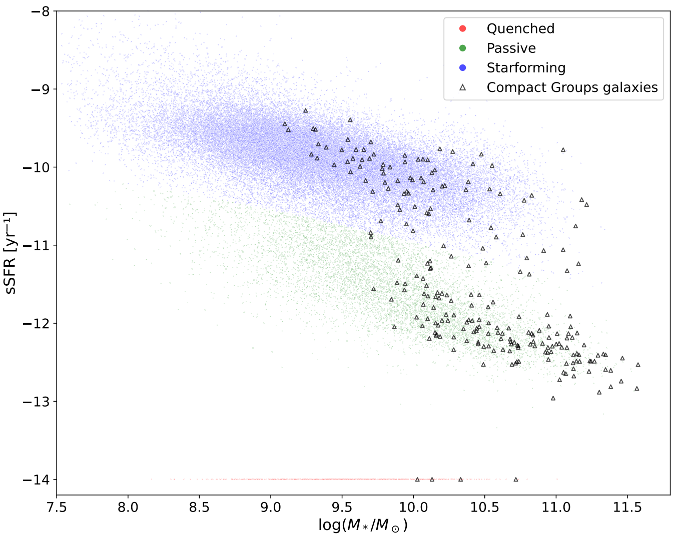

Fig. 5: sSFR and morphologies of galaxies in CG_4 and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0}$ Gyr^{-1}.

than 0.5, and "spiral" if the `p_cs_debiased` value is greater than 0.5. Other cases are considered as "uncertain". The morphologies found for each sample are shown in table 2. Barnard two-sided exact tests were performed to compare the fraction of each morphology between CG_4 and each control sample. The p-values found are for ellipticals and for spirals versus $Control_{4B}$ sample, and versus $Control_{4C}$ and and versus RG_4. Morphologies of CG_4, wich we identified as cores of richer groups in Tricottet et al. (2025), have morphologies significantly different from ordinary groups of four members. Figure 5 displays the sSFR against masses for galaxies from the SDSS selection and for CG_4s galaxies.

2.4. Morphology vs sSFR

Starforming CG_4 are composed of of their galaxies classified as spiral (% once galaxies having uncertain morphology removed) and (% once galaxies having uncertain morphology removed) elliptical. These figures are (%) spiral and (%) elliptical galaxies for $Control_{4B}$, (%) spiral and (%) elliptical galaxies for $Control_{4C}$ and (%) spiral and (%) elliptical galaxies for RG_4. The probability that those fractions come from the same distribution is estimated from the as against $Control_{4B}$, against $Control_{4C}$ and against RG_4. There is thus neither significant excess nor lack of starforming elliptical galaxies in CG_4 compared to all our control samples.

If we restrict the comparison to BGGs only, we find Elliptical BGGs (% of the BGGs) and Spiral BGGs (%) in CG_4. This is to be compared to (%) Elliptical and (%) Spiral galaxies in $Control_{4B}$, (%) Elliptical and (%) Spiral in $Control_{4C}$, and (%) Elliptical and (%) Quenched in RG_4. The p-values found from are against $Control_{4B}$, against $Control_{4C}$ and against RG_4. There is thus no significant difference in the fraction of star-forming BGGs between CG_4 and our control samples.

3. sSFR

3.1. Main sequence

We model the star-forming main sequence as a power law in stellar mass, to various orders (1 to 4) - cf figure . We then compute the residuals of each star-forming galaxy to the main sequence,and plot their histograms in figure .

The median main-sequence residual of CG4 galaxies is significantly different
from that of Control4B, with a median offset of $\Delta = 0.057$ ($0.028 \leq \Delta \leq 0.099$, bootstrap), corresponding to a p-value of $p = 0.056$.

The median main-sequence residual of CG4 galaxies is significantly different
from that of Control4C, with a median offset of $\Delta = 0.082$ ($0.062 \leq \Delta \leq 0.122$, bootstrap), corresponding to a p-value of $p = 0.011$.

The median main-sequence residual of CG4 galaxies is not significantly different
from that of RG4, with a median offset of $\Delta = 0.046$ ($-0.037 \leq \Delta \leq 0.098$, bootstrap), corresponding to a p-value of $p = 0.533$.

Sample	**Elliptical**	**Spiral**	**Uncertain**
CG_4	52 (%)	53 (%)	143 (%)
$Control_{4B}$	289 (%)	583 (%)	1924 (%)
$Control_{4C}$	533 (%)	781 (%)	1694 (%)
RG_4	17 (%)	75 (%)	132 (%)
SDSS selection	*10091 (%)*	*34715 (%)*	*7725 (%)*

Table 2: Number of galaxies of each morphology for each sample.

Sample	$\langle \log sSFR \rangle$		$\langle M_r \rangle$		$\langle \log(M_\star/M_\odot) \rangle$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	−12.15	−10.15	−21.04	−20.18	10.65	9.99
Control4B	−12.18	−10.28	−21.68	−20.71	10.87	10.27
Control4C	−12	−10.14	−20.98	−19.71	10.55	9.75
RG4	−11.92	−10.1	−21.31	−20.12	10.76	9.94

Table 3: Median values of galaxy properties for passive and star-forming populations in each sample. Error bars are computed via bootstrapping (10 000 iterations).

Sample	sSFR (yr^{-1})		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	−11.99 ± 0.09	−10.75 ± 0.29	−20.84 ± 0.11	−20.66 ± 0.32	10.46 ± 0.06	10.28 ± 0.12
Control4B	−11.92 ± 0.03	−10.7 ± 0.06	−21.32 ± 0.03	−21.11 ± 0.07	10.69 ± 0.02	10.54 ± 0.04
Control4C	−11.53 ± 0.03	−10.5 ± 0.06	−20.27 ± 0.05	−20.56 ± 0.08	10.18 ± 0.02	10.24 ± 0.06
RG4	−11.36 ± 0.19	−10.39 ± 0.16	−20.69 ± 0.14	−20.51 ± 0.27	10.3 ± 0.07	10.18 ± 0.12

Table 4: Median values of various quantities for galaxies according to the fertility of their BGG (BGG included). Error bars are computed via bootstrapping (10 000 iterations).

Sample	sSFR (yr^{-1})		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	−11.56 ± 0.16	−10.84 ± 0.47	−20.41 ± 0.13	−20.28 ± 0.18	10.2 ± 0.05	10.12 ± 0.06
Control4B	−11.46 ± 0.06	−10.68 ± 0.13	−20.98 ± 0.03	−20.77 ± 0.06	10.49 ± 0.02	10.37 ± 0.03
Control4C	−11 ± 0.04	−10.37 ± 0.07	−19.72 ± 0.04	−20.04 ± 0.09	9.9 ± 0.02	9.9 ± 0.07
RG4	−10.64 ± 0.19	−10.31 ± 0.22	−20.15 ± 0.16	−20.24 ± 0.13	10.08 ± 0.08	10.06 ± 0.13

Table 5: Median values of various quantities for galaxies according to the fertility of their BGG (satellites only). Error bars are computed via bootstrapping (10 000 iterations).

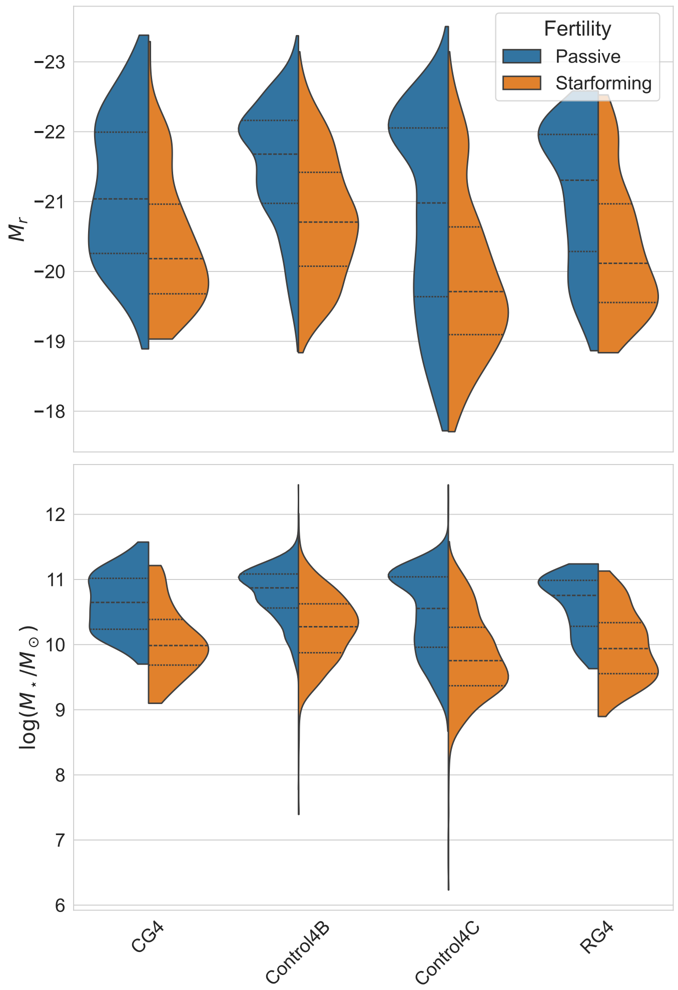

Fig. 6: Distribution various quantities for galaxies according their fertility.

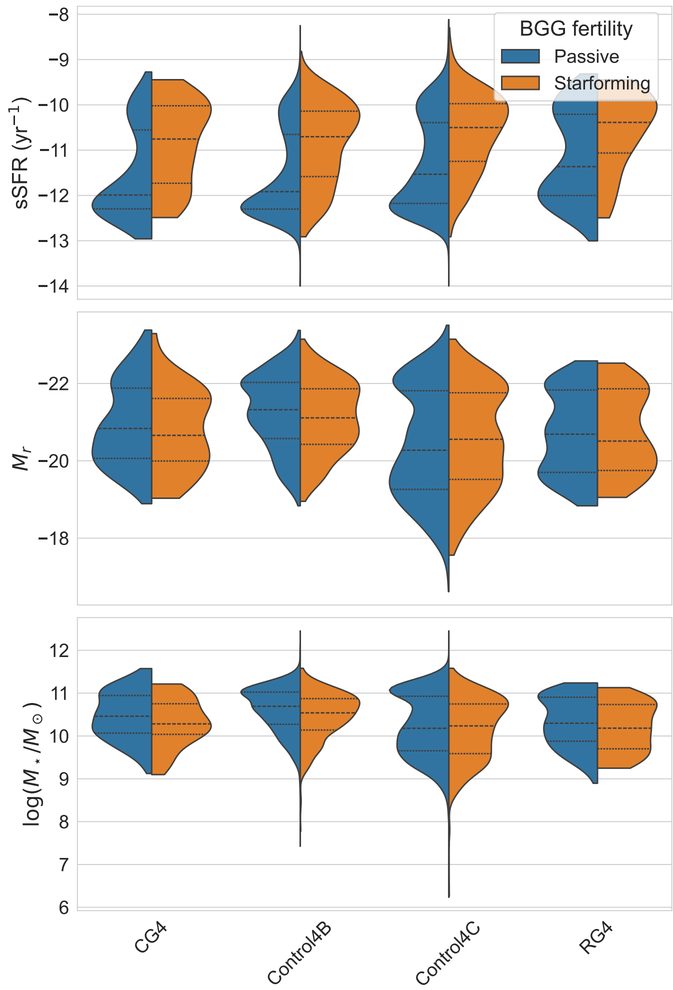

Fig. 7: Distribution various quantities for galaxies according to the fertility of their BGG (BGG included).

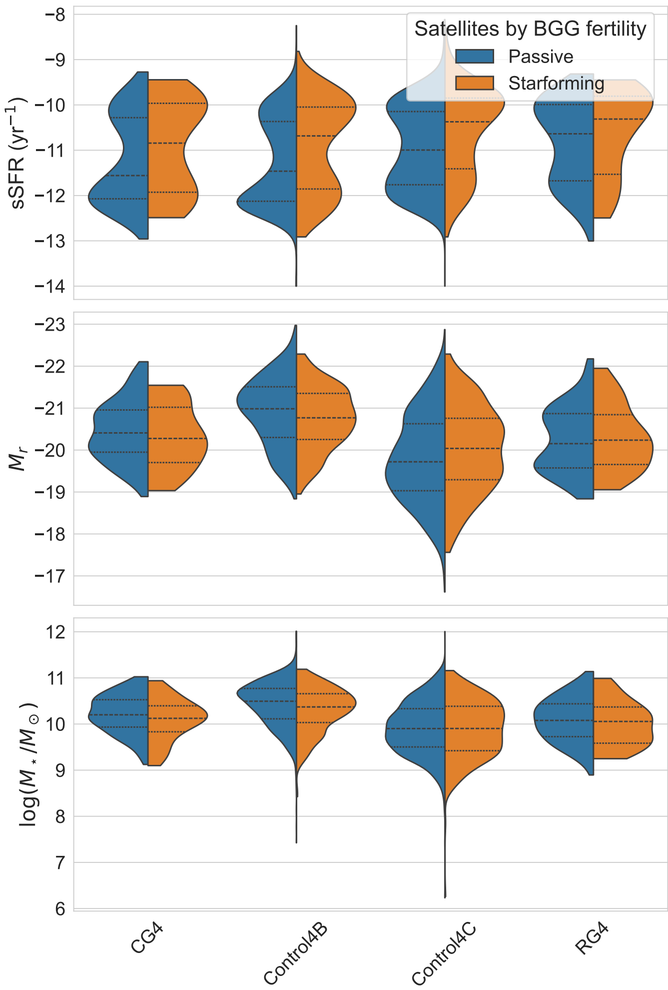

Fig. 8: Distribution various quantities for galaxies according to the fertility of their BGG (BGG excluded).

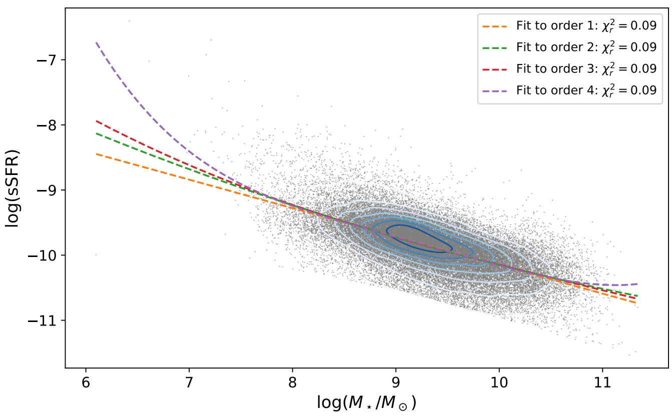

Fig. 9

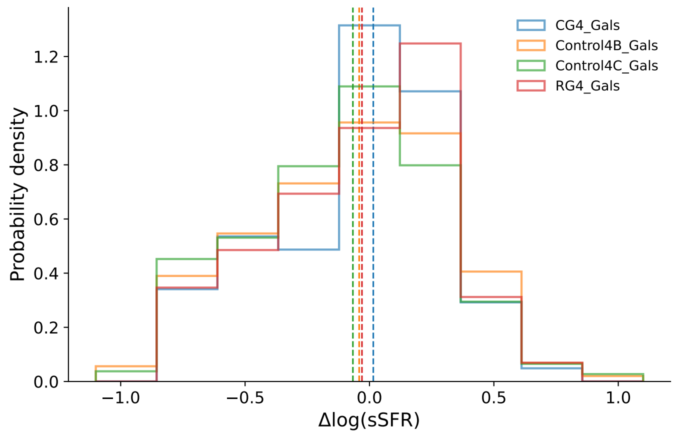

Fig. 10

Interpolation points for sSFR classification

Acknowledgements. We thank ...

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